Realiza las siguientes operaciones en R deber´a adjuntar tanto el c´odigo utilizado

como los resultados obtenidos.

1 Efectu´a las siguientes operaciones aritm´eticas

- 7-(-8+9)-9
  [1] -3
- 1-(5*3-5)

  [1] -9

- 18.99-2.5*(1.79-24.28)

  [1] 75.215

- (-3)^5 * (+2)^4 * ((-4)^2 + (-2))
  [1] -54432
- ((-3)^5 * (+7)^2) / ((-4)^2 + (-2))
  [1] -850.5

2 Dadas las matrices

```
A <- matrix(
+   c(2, 0, 1,
+     3, 0, 0,
+     5, 1, 1),
+   nrow = 3,
+   byrow = TRUE
+ )
>
> B <- matrix(
+   c(1, 0, 1,
+     1, 2, 1,
+     1, 1, 0),
+   nrow = 3,
+   byrow = TRUE
+ )
A + B
     [,1] [,2] [,3]
[1,]    3    0    2
[2,]    4    2    1
[3,]    6    2    1
> A - B
     [,1] [,2] [,3]
[1,]    1    0    0
[2,]    2   -2   -1
```

```
[3,]    4    0    1
> A%*%B
     [,1] [,2] [,3]
[1,]    3    1    2
[2,]    3    0    3
[3,]    7    3    6
> B%*%A
     [,1] [,2] [,3]
[1,]    7    1    2
[2,]   13    1    2
[3,]    5    0    1
```

Demostrar que $A^2 - A - 2I = 0$

```
A <- matrix(
+   c(0, 1, 1,
+     1, 0, 1,
+     1, 1, 0),
+   nrow = 3,
+   byrow = TRUE
+ )
>
> I <- matrix(
+   c(1, 0, 0,
+     0, 1, 0,
+     0, 0, 1),
+   nrow = 3,
+   byrow = TRUE
+ )
```

```
Acuad <- A %*% A
>
> Acuad_A <- Acuad - A
>
> resultado <- Acuad - A - 2 * I
>
> A
     [,1] [,2] [,3]
[1,]    0    1    1
[2,]    1    0    1
[3,]    1    1    0
> I
```

```
     [,1] [,2] [,3]
[1,]    1    0    0
[2,]    0    1    0
[3,]    0    0    1
> Acuad
     [,1] [,2] [,3]
[1,]    2    1    1
[2,]    1    2    1
[3,]    1    1    2
> Acuad_A
     [,1] [,2] [,3]
[1,]    2    0    0
[2,]    0    2    0
[3,]    0    0    2
> resultado
     [,1] [,2] [,3]
[1,]    0    0    0
[2,]    0    0    0
[3,]    0    0    0
```